ELE201 Semester Project Report

Dual-Technology Distance Measurement: Ultrasonic (HC-SR04) and Infrared (Sharp GP2Y0A21YK0F)

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Abstract

This project compares an HC-SR04 ultrasonic sensor and a Sharp GP2Y0A21YK0F infrared (IR) sensor measuring the same target with an STM32F767ZI Nucleo-144 board. Distances from 10 to 95 cm were measured on three surfaces (white paper, dark jacket, aluminium foil), with 30 samples per point, plus bright light and a tilted target. The ultrasonic sensor had a small, nearly constant bias (mostly -1 to -2.5 cm) on hard surfaces over the whole range, but failed on a tilted target. After calibration, the IR sensor was within about ± 3 cm up to 80 cm on paper and fabric, but failed on foil (down to -10 cm) and had large errors at 95 cm because its output curve flattens. The distance is shown on an LCD, and three LEDs and a buzzer with rising pitch give a proximity alert.

1 Methodology

1.1 Tools and components

Hardware:

- Microcontroller: STM32F767ZI Nucleo-144
- IR sensor: Sharp GP2Y0A21YK0F [2]
- Ultrasonic sensor: HC-SR04 [1]
- LCD: TC1602A-21T [3]
- 3 LEDs with 220 Ω resistors
- Passive piezo buzzer, PKM22EPP-40xx series [4]
- 100 µF supply capacitor for the IR sensor
- TMP36 for air temperature

Software:

- STM32CubeMX (HAL configuration)
- VS Code with PlatformIO (code and upload)
- KiCad (schematic)
- PuTTY (serial logging)
- Spreadsheet (analysis of the measurements)
- Overleaf (report)

1.2 Concepts used

- **Clock and timers:** 16 MHz internal oscillator and PLL to 108 MHz; all timers use a prescaler giving 1 µs per tick. TIM4 channel 4 is PWM that generates the 10 µs trigger pulse every 60 ms. TIM1 channel 1 is PWM for the buzzer.
- **Interrupts / input capture:** TIM4 channel 3 captures the timer value in hardware on each ECHO edge. The interrupt switches between rising and falling edge, and the pulse width is the difference of the two captures.
- **Analog input:** 12-bit ADC (PA3), average of 16 readings 3 ms apart.
- **Digital I/O:** LCD in 4-bit mode and three LED outputs. **Serial I/O:** USART3, 115200 baud, CSV logging to a PC.
- **Alert:** buzzer sounds below 50 cm with frequency $f = 4500 - 30\,d$ (Hz, d in cm); LEDs are green above 30 cm, yellow 16–30 cm and red at 15 cm or less.

1.3 How the sensors measure

The ultrasonic module sends an 8-cycle 40 kHz burst after the trigger pulse and holds ECHO high until the echo returns. The distance is

$$d_{\mathrm{US}} = \frac{c\,t}{2}, \qquad c = 331.3 + 0.606\,T \ [\mathrm{m/s}] \quad \Rightarrow \quad d_{\mathrm{US}}\,[\mathrm{cm}] \approx \frac{t\,[\mu\mathrm{s}]}{58}, \tag{1}$$

with the factor 2 because the sound travels there and back. ECHO is 5 V, so a divider gives $V = 5\,\mathrm{V} \cdot 2/3 \approx 3.3\,\mathrm{V}$ on the pin.

The IR sensor works by triangulation: a lens focuses the reflected light on a position-sensitive detector, and the position of the light spot, which shifts with distance, is converted to an analog voltage that is roughly proportional to $1/d$. It is read by the ADC as $V = \mathrm{ADC} \cdot 3300/4095$ mV and converted by

$$d_{\mathrm{IR}}\,[\mathrm{cm}] = \frac{a}{V - b}, \qquad a = 26.18\ \mathrm{cm \cdot V},\ b = 0.074\ \mathrm{V}. \tag{2}$$

a and b were fitted by least squares on 1/d versus V for white paper at 20–95 cm, then tested on the other surfaces. The LCD shows the IR value only inside a trusted voltage range (about 10–85 cm) and otherwise reports out of range.

1.4 System overview, communication and flowchart

The sensors send signals to the MCU (ECHO to a timer input, IR voltage to the ADC). The MCU drives the LCD (4-bit GPIO), the LEDs (GPIO) and the buzzer (PWM), and sends raw data to the PC over UART. The full circuit is in the appendix.

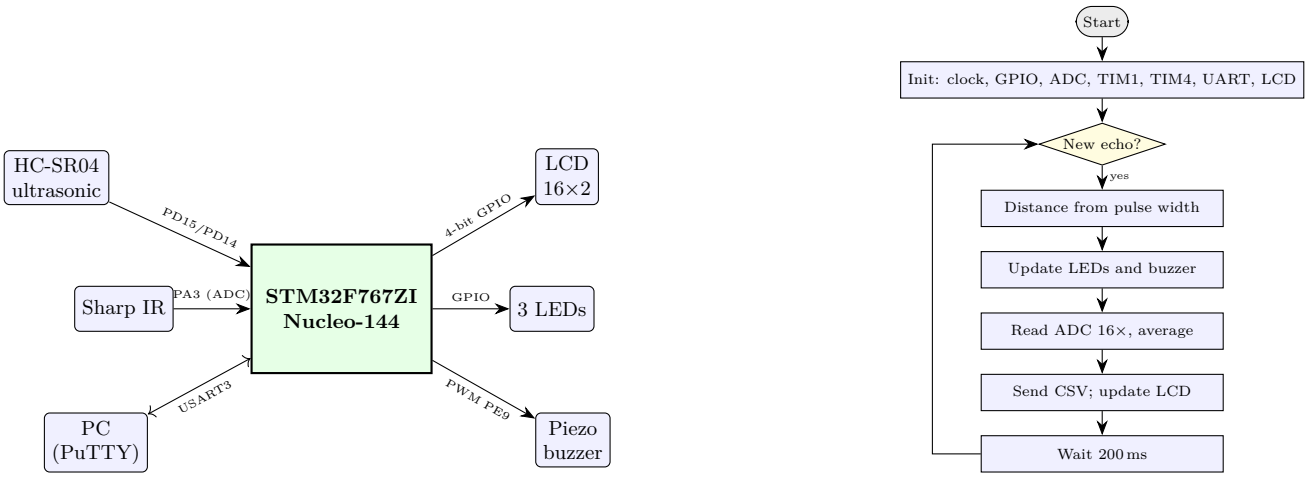


Figure 1: Block diagram (left) and flowchart of the main loop (right). In the echo-capture interrupt, the rising edge stores the time and the falling edge computes the pulse width and sets a flag.

1.5 Measurement procedure

Targets (white paper, an Arc'teryx dark jacket, aluminium foil) were placed at 10, 20, 40, 60, 80 and 95 cm from a tape-measure zero, with 30 samples per point and a reset between distances. The sensors were fixed on a platform; the transducers protrude about 2 cm, so the zero point is approximate. Air temperature was 21.4 °C (TMP36). Extra tests: bright light on white paper (10, 20, 40 cm) and white paper tilted about 30° (10, 20, 40 cm; the angle was set by hand). Raw pulse width and ADC average were logged as CSV, and mean, standard deviation (SD) and bias (mean minus true) were computed in a spreadsheet.

2 Results

Table 1: White paper: mean, SD and bias (all in cm) for 30 samples per distance.

	Ultrasonic			IR (calibrated)		
True	Mean	SD	Bias	Mean	SD	Bias
10	8.8	0.12	−1.2	10.8	0.01	+0.8
20	18.4	0.02	−1.6	20.1	0.06	+0.1
40	38.1	0.22	−1.9	39.0	0.40	−1.0
60	58.1	0.34	−1.9	57.3	1.15	−2.7
80	77.9	0.54	−2.1	82.9	1.18	+2.9
95	93.1	0.80	−1.9	100.4	2.77	+5.4

Table 2: Dark jacket: mean, SD and bias (all in cm) for 30 samples per distance.

	Ultrasonic			IR (calibrated)		
True	Mean	SD	Bias	Mean	SD	Bias
10	11.8	1.03	+1.8	12.6	0.17	+2.6
20	20.9	0.25	+0.9	21.1	0.06	+1.1
40	41.8	1.50	+1.8	39.4	0.51	−0.6
60	60.4	0.47	+0.4	60.0	0.44	+0.0
80	79.0	0.45	−1.0	82.7	1.33	+2.7
95	94.4	0.76	−0.6	102.5	1.86	+7.5

Table 3: Aluminium foil: mean, SD and bias (all in cm) for 30 samples per distance.

	Ultrasonic			IR (calibrated)		
True	Mean	SD	Bias	Mean	SD	Bias
10	9.5	0.13	−0.5	11.5	0.01	+1.5
20	20.0	0.36	0.0	20.7	0.06	+0.7
40	38.3	0.24	−1.7	32.1	0.33	−7.9
60	58.2	0.35	−1.8	49.8	0.82	−10.2
80	77.5	0.46	−2.5	73.2	0.78	−6.8
95	93.6	0.89	−1.4	100.1	20.02	+5.1

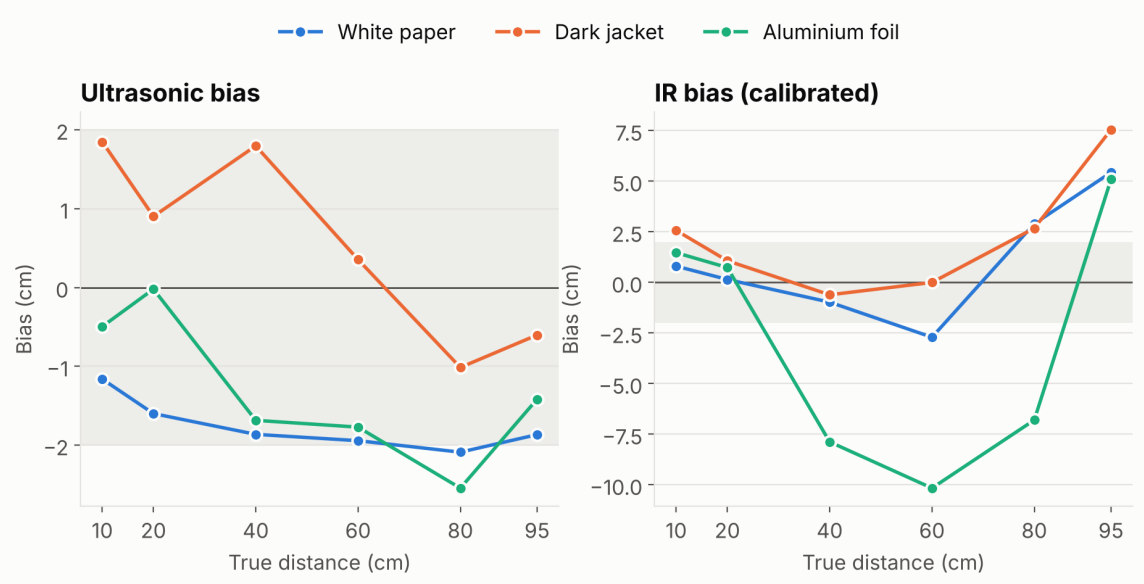


Figure 2: Bias versus true distance for each surface. The grey band is ±2 cm. Note the different vertical scales.

Ultrasonic. On hard, flat surfaces the bias was mostly −1 to −2.5 cm (foil: −0.5 and 0.0 cm at 10 and 20 cm) with SD below 0.9 cm, so the error looks like a zero-point offset rather than a scale error. A likely cause is that the

transducers protrude about 2 cm from the platform, but this was not tested separately. Compared with paper, the dark jacket shifted the result by 1–4 cm and increased the SD (1.0 cm at 10 cm versus 0.12 cm on paper), probably because soft fabric reflects sound less cleanly. The temperature correction was small (about 0.16 % at 21.4 °C).

IR. Within 10–40 cm the IR error was about 1 cm or less on paper and within 2.6 cm on jacket. Beyond 60 cm the output curve is so flat (about 5 mV per cm) that a 50 mV error gives about 10 cm; this is a property of the $1/d$ sensor curve and not of the surface, since paper and jacket gave almost the same voltage at 80 and 95 cm. At 95 cm on foil the IR reading was unstable (SD 20 cm). Calibrating on paper and testing on the jacket gave similar errors (80 cm: +2.9 and +2.7 cm). On aluminium foil the IR error was up to −10 cm: a mirror reflects the light in one direction instead of scattering it, which the datasheet warns about. The measured output at 80 cm was 0.390 V (datasheet: 0.25–0.55 V, typical 0.4 V), and the difference between 10 and 80 cm was 2.11 V (datasheet: 1.65–2.15 V), so the sensor was within specification [2] (Figure 3, left).

Conditions. Bright light changed the IR voltage by 6–126 mV, but the ultrasonic sensor (unaffected by light) showed that the target had moved by about 0.5 cm, which explains the difference, so no light effect was detected. With a tilted target the ultrasonic sensor gave 83 and 97 cm at true 10 and 20 cm (the echo was reflected away), but 38 cm at 40 cm; IR gave 20.8 and 41.5 cm at 20 and 40 cm (Figure 3, right). The 10 cm tilt value for IR was unstable (hand-held angle).

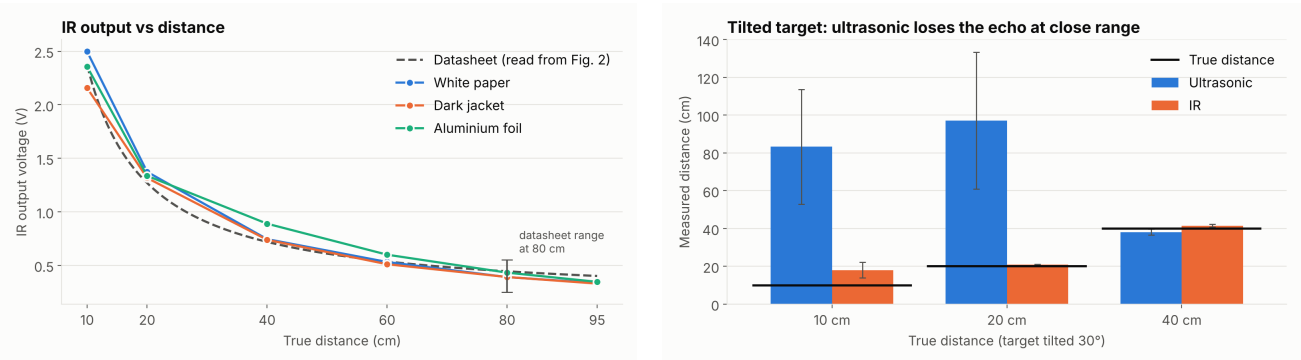


Figure 3: Left: IR output voltage versus distance for three surfaces, with the datasheet curve (read from the datasheet figure, about ±0.05 V). Right: tilted white paper, measured versus true distance (bars show SD).

3 Challenges

- **The biggest challenge:** definitely the IR readings. In the beginning I never got constant readings within an acceptable range. I first thought it was because I had not calibrated it. Then I used my multimeter and measured the GND difference between the Nucleo and the Sharp sensor. It showed a difference of 0.47 V, which explains my odd measurements. My biggest suspicion then turned to the connections. I used a JST connector with thin JST crimp wires, and these fell out of their contacts several times. I was also sceptical of the connection between the jumper cable and the crimp wire, so I soldered them together. I also moved the GND and 5 V cables directly to the bus on the breadboard and removed one unnecessary GND cable. After this the difference fell to 0.01–0.04 V.
- **Expected difficult, but went well:** the LCD integration. I had not done this before and thought it would be difficult. It worked on the first try and did not cause any problems. The only issue was that it showed gibberish when I measured with the multimeter, but a reset fixed it.
- **Expected easy, but tricky:** the data collection. As explained, I struggled a lot with the IR measurement. I also struggled to set up a working test station. I am aware of the limitations of the test station and that it gives a lot of uncertainty in the data.

4 Conclusion

- **Demonstration:** see video part two. It demonstrates the LCD display, how the alert system works and, to some degree, the test station used for data collection.
- **Summary of results:** the ultrasonic sensor is accurate over the full range on hard surfaces but sensitive to tilt and soft surfaces. The IR sensor is robust to tilt and fabric, accurate to about 1–2 cm up to 40 cm and within about ±3 cm up to 80 cm on paper and fabric, but it fails on mirror-like surfaces and has large errors at 95 cm. They fail in different places, so they complement each other.
- **Last thoughts:** these results confirm the theory behind the two sensors. They must be seen in the context of the uncertainty of the test station. A fixed mount for the sensors and a bigger, stable target with more

accurate placement should have been in place. I believe human placement errors explain several of the differences in this report, and this should not have been a factor. In a more professional setting where this would have been possible, the project would give a very good picture of the two sensors.

AI declaration

I used Claude throughout as a tutor and an assistant. It helped me organise the whole project. I asked it to use a Socratic question format: it asked me questions about the project, I answered as well as I could, and it responded with more questions. This was helpful because I wrote the answers myself, but still received guidance through its questions. The area where I used AI the most was data collection. It created my spreadsheet with all the formulas, so I could paste in the data and the rest, including the graphs, filled out automatically. It also helped me create the CSV logging. The data collection would not have been as detailed without it. It also helped me quite a bit with setting up the LCD physically and with the code, and for these parts it was extremely helpful. Claude was also used for the skeleton of this report and for helping me with Overleaf, which I had never used before. On the other hand, I tried to use it for troubleshooting the odd IR measurements, but it made things more confusing and did not help much. This kind of problem is not well suited to AI, because it cannot see what is actually on the breadboard. Only a person can troubleshoot that properly, and for that you need to know and understand what you are doing. It was also a comfort to have AI since I worked individually.

References

[1] Multicomp, *Ultrasonic Ranging Module HC-SR04, datasheet*, via alldatasheet.com, accessed 7 November 2026.

[2] Sharp Corporation, *GP2Y0A21YK0F Distance Measuring Sensor Unit, datasheet*, Sheet No. E4-A00201EN, 1 December 2006.

[3] Tinsharp Electronics, *TC1602A-21T 16×2 LCD module, drawing*, 2007.

[4] Murata Manufacturing, *Piezoelectric Sounders, External Drive Pin Type (PKM22EPP series)*, catalog P37E.

[5] Anthropic, *Claude Sonnet 5.5*

Appendix

Code and spreadsheet: `https://github.com/daviknes-sudo/Semesterprosjekt_ELE201_Daviknes`

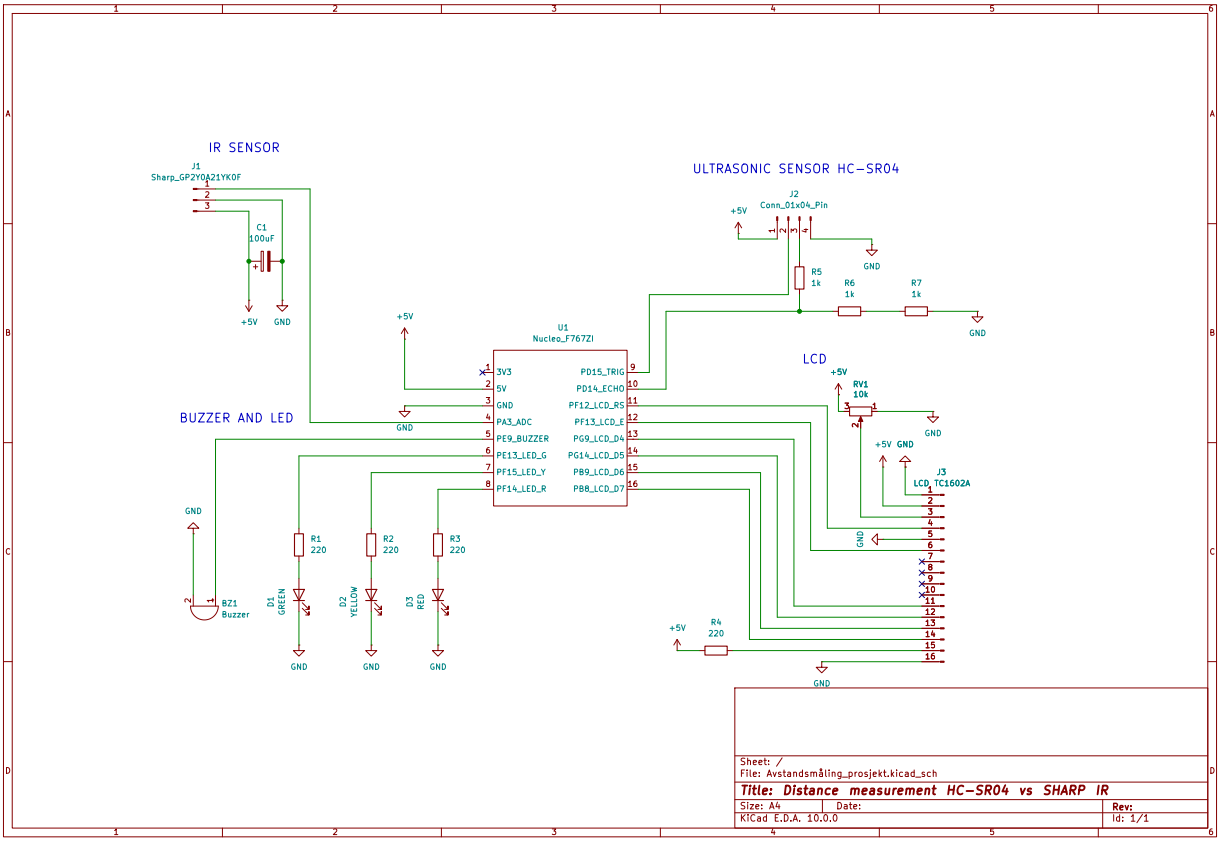


Figure 4: Circuit schematic (KiCad).

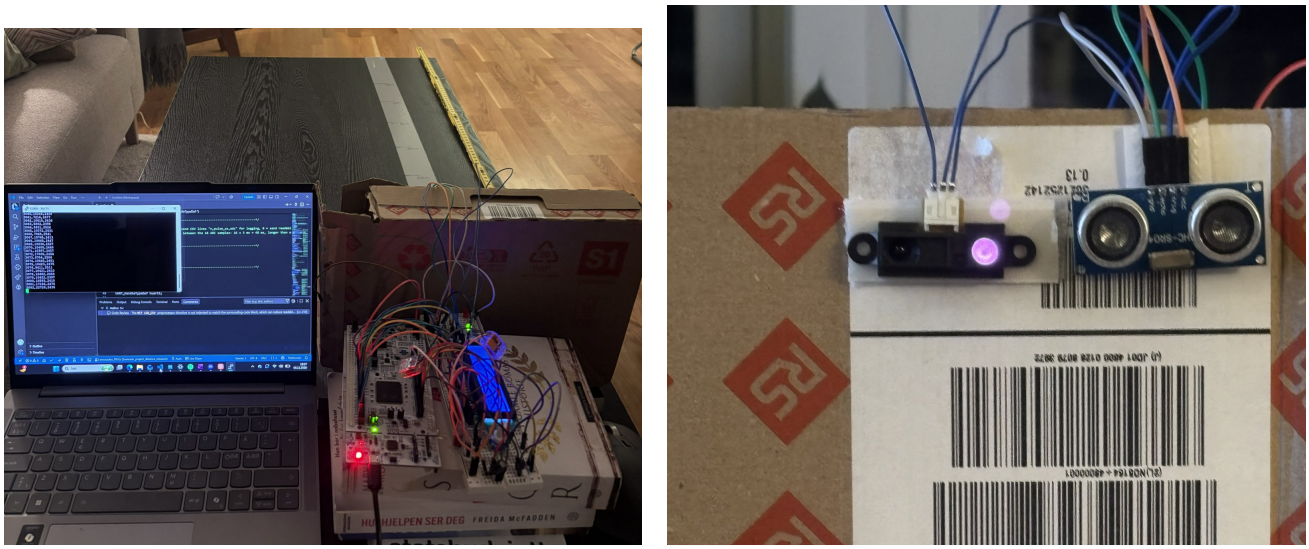


Figure 5: Test setup: overview (left) and close-up of the sensors (right).